

Computer Vision SS26 – Exercise Sheet

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This exercise will be discussed the week of 6th of July.

1 Introduction

The following exercises cover surface reconstruction, point clouds, image classification, nearest-neighbor classifiers, and linear classifiers.

2 Exercises

Exercise 1: From Depth Maps to 3D Reconstruction

A stereo reconstruction pipeline produces depth maps from dense correspondences. These depth maps are then fused into a 3D reconstruction.

1. Name the main steps of the 3D reconstruction pipeline, starting from input images and ending with a 3D model.
2. Explain why depth maps from several camera views need to be fused.
3. Suppose a calibrated camera observes a pixel (u, v) with depth z . The camera intrinsics are

$$K = \begin{pmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}.$$

Derive the corresponding 3D point (X, Y, Z) in camera coordinates.

4. Given $f_x = f_y = 500$, $c_x = 320$, $c_y = 240$, and a pixel $(u, v) = (420, 290)$ with depth $z = 2$, compute the 3D point.

Exercise 2: Point Clouds and Surface Reconstruction

A point cloud is a set of 3D points sampled from the surface of an object.

1. What information is contained in a point cloud? What information is missing compared to a triangle mesh?
2. Explain why point clouds are useful for digitizing real-world objects.
3. Consider the following five 2D points as a simplified point cloud:

$$p_1 = (0, 0), \quad p_2 = (1, 0), \quad p_3 = (1, 1), \quad p_4 = (0, 1), \quad p_5 = (0.5, 0.5).$$

Sketch or describe a reasonable triangulation.

4. Why can noise and missing measurements make surface reconstruction difficult?

Exercise 3: Nearest-Neighbor Image Classification

A nearest-neighbor classifier stores all training images and predicts the label of the most similar training image.

Consider the following training data with two-dimensional feature vectors:

Feature vector	Class
(1, 1)	cat
(2, 1)	cat
(5, 4)	dog
(6, 5)	dog
(1, 5)	toaster

A test image has feature vector

$$x = (3, 2).$$

1. Compute the Euclidean distance from x to every training example.
2. Which training example is the nearest neighbor?
3. What class is predicted?
4. For N training images and M test images, state the training and testing complexity of the nearest-neighbor classifier.

Exercise 4: Linear Image Classifier

A linear classifier assigns a score to each class using

$$s = Wx + b,$$

where x is the image feature vector, W is the weight matrix, and b is the bias vector.

Let

$$x = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad W = \begin{pmatrix} 1 & 0 \\ 0 & 2 \\ -1 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ -1 \\ 2 \end{pmatrix}.$$

The three rows of W correspond to the classes cat, dog, and toaster.

1. Compute the class score vector s .
2. Which class is predicted?
3. Explain the difference between a nearest-neighbor classifier and a linear classifier.
4. Why is fast testing often more important than fast training in image classification?